

Experiment 15

Custom Keywords and Libraries in Robot Framework

This experiment shows how Robot Framework can be extended with a Python library and how its BuiltIn library can evaluate a simple arithmetic expression. Both approaches add 15.5 and 4.5 and should produce 20.0.

Problem Statement

Create a Robot Framework test that performs an addition in two ways: first through a custom Python keyword, and then through the BuiltIn Evaluate keyword. Record the results in the Robot Framework log.

Objective

Learn how to connect a Python module to Robot Framework, expose a Python function as a test keyword, and use a standard BuiltIn keyword for calculations.

Software and Concepts

- Python 3.9 or newer
- Robot Framework 7.x
- Visual Studio Code
- Python modules and functions
- Robot Framework library imports
- BuiltIn Evaluate and logging keywords

Implementation

1 Create the Python library

Save the following code as CustomMath.py. The function converts both inputs to floating-point values before returning their sum.

```
def calculate_sum(num1, num2):  
    """Return the arithmetic sum of two values as a float."""  
    return float(num1) + float(num2)
```

2 Create the Robot Framework test

Save the following as Experiment3.robot. Importing CustomMath.py makes calculate_sum available as the Robot Framework keyword Calculate Sum.

```
*** Settings ***  
Library CustomMath.py  
Library BuiltIn
```

*** Test Cases ***

Custom Python Keyword Adds Two Values

```
${result}= Calculate Sum 15.5 4.5  
Should Be Equal As Numbers ${result} 20.0  
Log The custom Python sum is: ${result}
```

BuiltIn Evaluate Adds Two Values

```
${result}= Evaluate 15.5 + 4.5  
Should Be Equal As Numbers ${result} 20.0  
Log The BuiltIn sum is: ${result}
```

3 Install and run

Open the project folder in VS Code and run these commands in its terminal:

```
python -m venv .venv  
.\.venv\Scripts\Activate.ps1  
python -m pip install -r requirements.txt  
python -m robot --outputdir results Experiment3.robot
```

Robot Framework creates results\log.html and results\report.html. Open log.html in a browser to inspect the keyword calls and logged values.

Expected Results

Test case	Method	Expected value
Custom Python Keyword Adds Two Values	Calculate Sum	20.0
BuiltIn Evaluate Adds Two Values	Evaluate	20.0

Result and Observation

Both tests should pass. The Python function and the BuiltIn expression return the same numerical result, while the log records how each value was obtained. Robot Framework automatically exposes the Python function as a keyword with a readable, space-separated name.

```
Requirement already satisfied: robotframework<8.0,>=7.0 in .\venv\Lib\site-packages (from -r requirements.txt (line 1)) (7.5)  
Requirement already satisfied: python-docx<2.0,>=1.1 in .\venv\Lib\site-packages (from -r requirements.txt (line 2)) (1.2.0)  
Requirement already satisfied: lxml>=3.1.0 in .\venv\Lib\site-packages (from python-docx<2.0,>=1.1->-r requirements.txt (line 2)) (6.1.3)  
Requirement already satisfied: typing_extensions>=4.9.0 in .\venv\Lib\site-packages (from python-docx<2.0,>=1.1->-r requirements.txt (line 2)) (4.16.0)  
=====  
Experiment3  
=====  
Custom Python Keyword Adds Two Values :: Calls the user-defined Py... | PASS |  
-----  
BuiltIn Evaluate Adds Two Values :: Uses Robot Framework's BuiltIn... | PASS |  
-----  
Experiment3 | PASS |  
2 tests, 2 passed, 0 failed  
=====
```

Conclusion

Robot Framework supports standard operations through BuiltIn keywords and can be extended with Python whenever a project needs custom calculations, business rules, or integrations. This combination makes the test suite both convenient for common actions and flexible for specialized behavior.